

Unit:1

Introduction to Operating System

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History Of OS

- Operating systems were first developed in the late 1950s to manage tape storage
- The General Motors Research Lab implemented the first OS in the early 1950s for their IBM 701
- In the mid-1960s, operating systems started to use disks
- In the late 1960s, the first version of the Unix OS was developed
- The first OS built by Microsoft was DOS. It was built in 1981 by purchasing the 86-DOS software from a Seattle company
- The present-day popular OS Windows first came to existence in 1985 when a GUI was created and paired with MS-DOS.

Introduction of Operating System

- ✓ Operating system is a system software designed to run a computer's hardware and application programs.
- ✓ Operating system is a program that acts as an intermediary between a user of a computer and the computer hardware to provide an environment in which a user can execute programs.
- ✓ An operating system is a set of program that controls and coordinates the overall function of computer system.
- ✓ The operating system is responsible for security, ensuring that unauthorized users do not access the system.
- ✓ The primary goal of an operating system is thus to make the computer system convenient to use.

What is an Operating System?

- A program that acts as an intermediary between a user of a computer and the computer hardware
- Operating system goals:
 - Execute user programs and make solving user problems easier
 - Make the computer system convenient to use
 - Use the computer hardware in an efficient manner

Operating Systems

- All devices have an operating system
- Software which manages the hardware
- Creates the interface between the hardware and the user
- Functions performed by the operating systems:
 - Managing the hardware resources of the computer
 - Providing the user-interface components
 - Providing a platform for software developers to write applications
- Most popular are Microsoft Windows, Apple Mac OS, and Linux



Examples of Operating System with Market Share



Generations of Operating System

Operating system is divided into four generations, which are explained as follows –

First Generation (1945-1955):

- ✓ It is the beginning of the development of electronic computing systems which are substitutes for mechanical computing systems.
- ✓ Because of the drawbacks in mechanical computing systems like, the speed of humans to calculate is limited and humans can easily make mistakes.
- ✓ In this generation there is no operating system, so the computer system is given instructions which must be done directly.
- **Example** – Type of operating system and devices used is Plug Boards.

Second Generation (1955-1965):

- ✓ The Batch processing system was introduced in the second generation, where a job or a task that can be done in a series, and then executed sequentially.
- ✓ In this generation, the computer system is not equipped with an operating system, but several operating system functions exist like FMS and IBSYS.
- **Example** – Type of operating system and devices used is Batch systems.

Third Generation (1965-1980):

- ✓ The development of the operating system was developed to serve multiple users at once in the third generation. Here the interactive users can communicate through an online terminal to a computer, so the operating system becomes multi-user and multiprogramming.
- **Example** – Type of operating system and devices used is Multiprogramming.

Fourth Generation (1980-Now):

- In this generation the operating system is used for computer networks where users are aware of the existence of computers that are connected to one another.
- At this generation users are also comforted with a Graphical User Interface (GUI), which is an extremely comfortable graphical computer interface, and the era of distributed computing has also begun.
- With the occurrence of new wearable devices like Smart Watches, Smart Glasses, VRGears, and others, the demand for conventional operating systems has also increased.
- And, with the onset of new devices like wearable devices, which includes Smart Watches, Smart Glasses, VR gears etc, the demand for unconventional operating systems is also rising.
- **Example** – Type of operating system and devices used is personal computers

Features of Operating Systems

- Here is a list of some important features of operating systems:
 - Provides a platform for running applications
 - Handles memory management and CPU scheduling
 - Provides file system abstraction
 - Provides networking support
 - Provides security features
 - Provides user interface
 - Provides utilities and system services
 - Supports application development

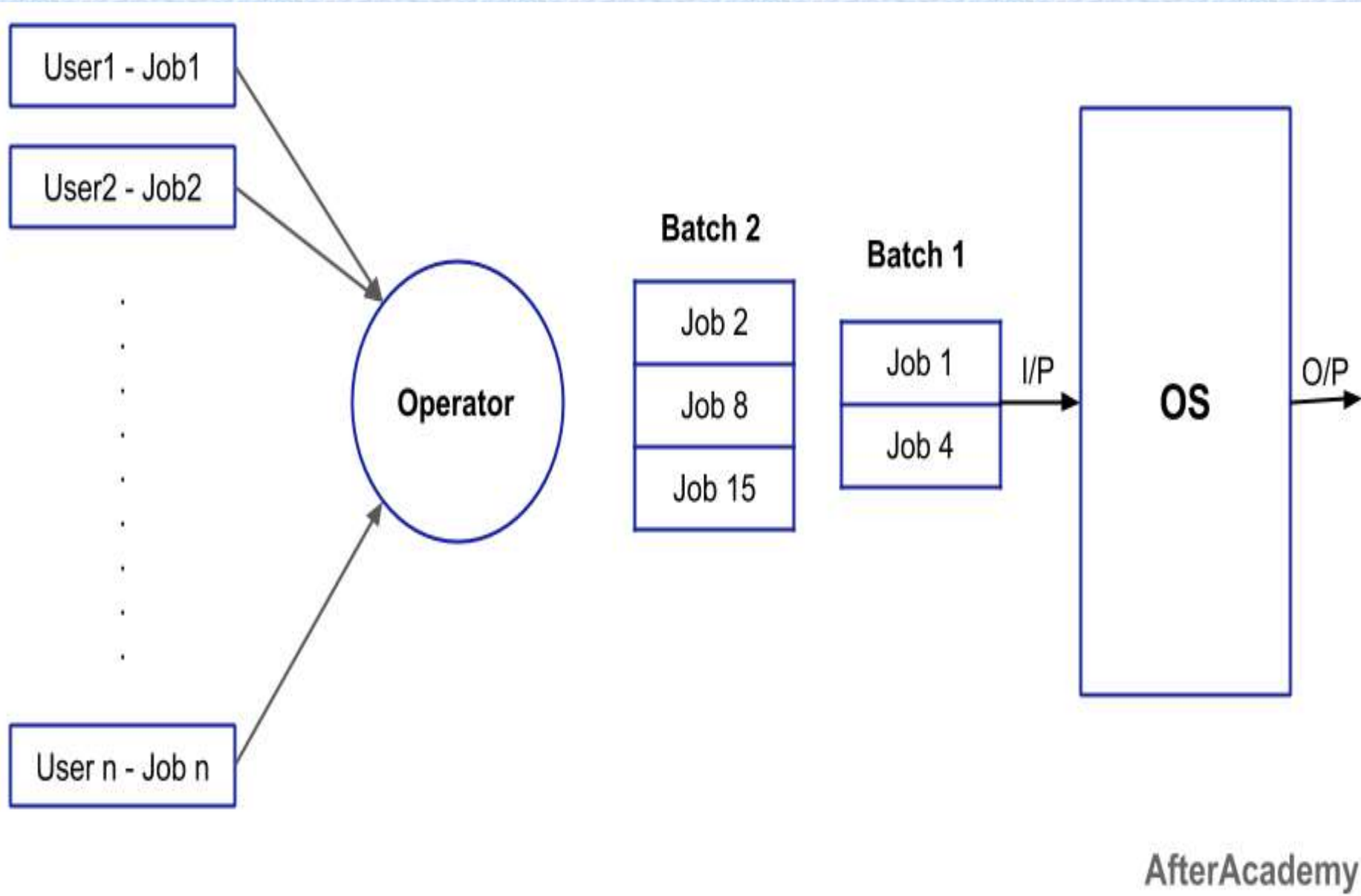
OS as an Extended Machine

- An operating system is the most important program in a computer system. This is one program that runs all the time, as long as the computer is operational and exits only when the computer is shut down.
- The fundamental goal of computer systems is to execute user programs and to make solving user problems easier. Hardware of a computer is equipped with extremely capable resources – memory, CPU, I/O devices etc.
- All these hardware units interact with each other in a well-defined manner. Also various programs require certain common operations, such as those controlling the I/O devices.
- The common functions of controlling and allocating resources are then brought together into one piece of software: the operating system.

What are the types of an Operating System?

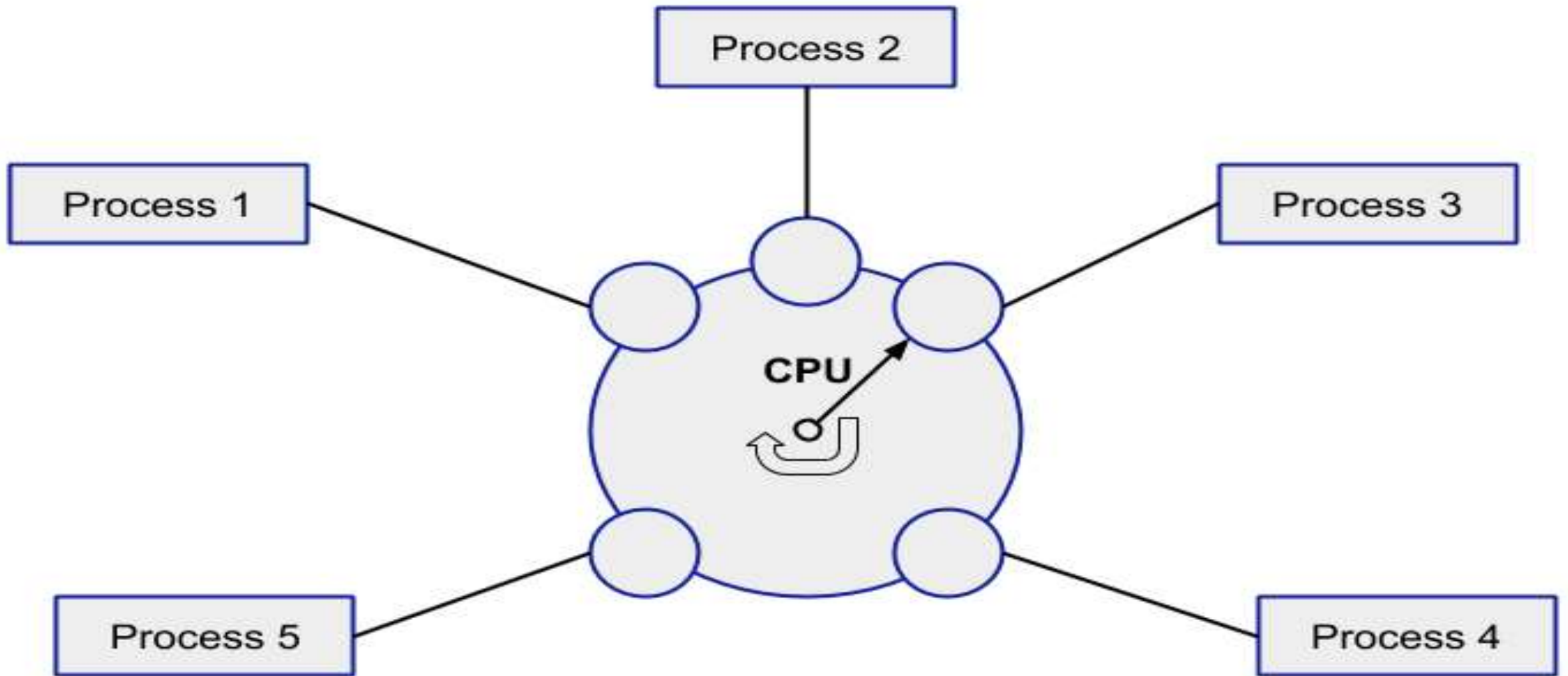
1. Batch Operating System:

- ✓ In a Batch Operating System, the similar jobs are grouped together into batches with the help of some operator and these batches are executed one by one.
- ✓ For example, let us assume that we have 10 programs that need to be executed. Some programs are written in C++, some in C and rest in Java.
- ✓ Now, every time when we run these programmes individually then we will have to load the compiler of that particular language and then execute the code.
- ✓ But what if we make a batch of these 10 programmes. The benefit with this approach is that, for the C++ batch, you need to load the compiler only once. Similarly, for Java and C, the compiler needs to be loaded only once and the whole batch gets executed.
- ✓ The following image describes the working of a Batch Operating System.



2. Time-Sharing Operating System

- ✓ In a Multi-tasking Operating System, more than one processes are being executed at a particular time with the help of the time-sharing concept.
- ✓ So, in the time-sharing environment, we decide a time that is called time quantum and when the process starts its execution then the execution continues for only that amount of time and after that, other processes will be given chance for that amount of time only.
- ✓ In the next cycle, the first process will again come for its execution and it will be executed for that time quantum only and again next process will come. This process will continue.
- ✓ The following image describes the working of a Time-Sharing Operating System.



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- **Advantages:**

1. Since equal time quantum is given to each process, so each process gets equal opportunity to execute.
2. The CPU will be busy in most of the cases and this is good to have case.

- **Disadvantages:**

1. Process having higher priority will not get the chance to be executed first because the equal opportunity is given to each process.

Advantages:

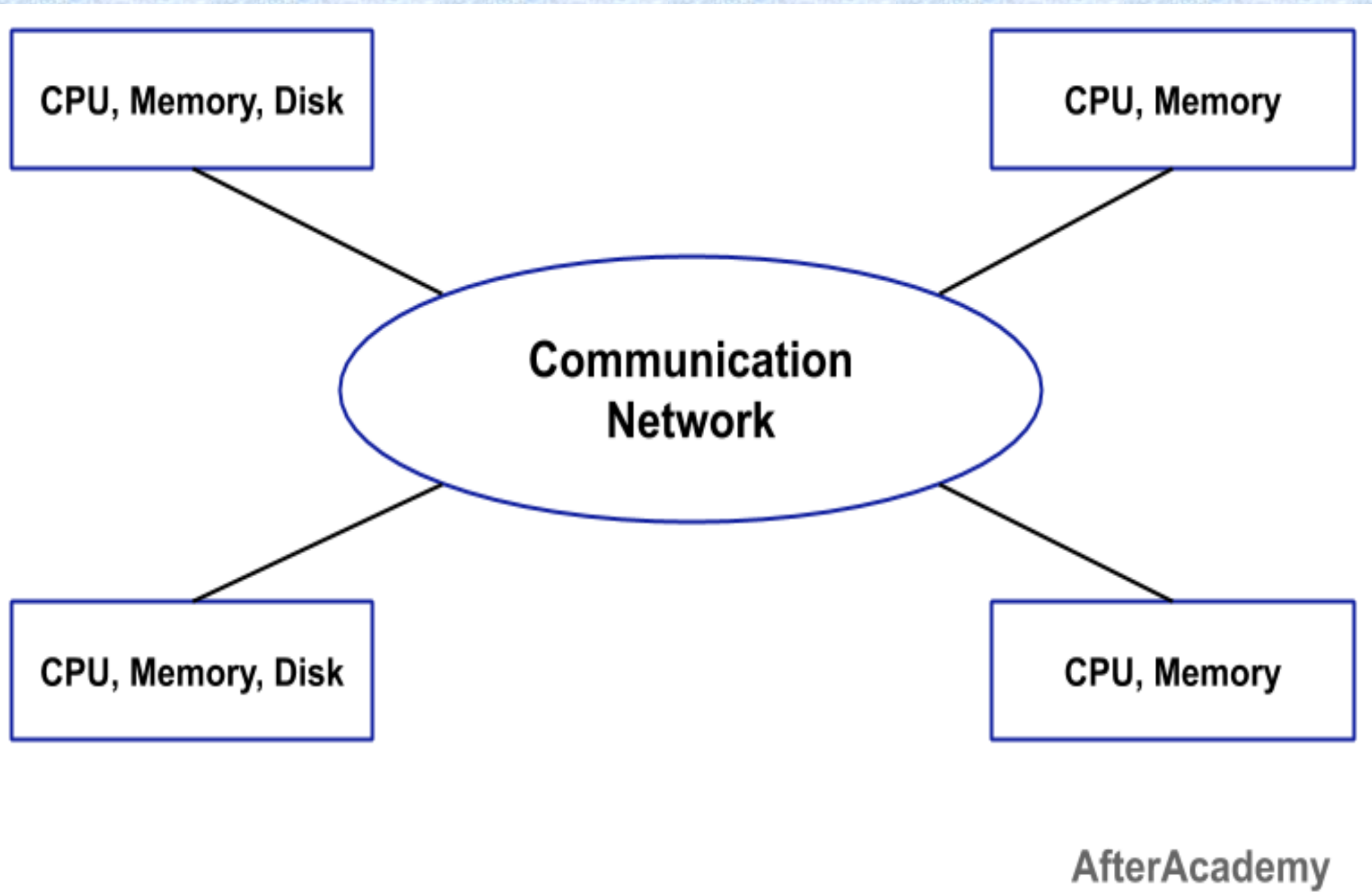
- 1.The overall time taken by the system to execute all the programmes will be reduced.
- 2.The Batch Operating System can be shared between multiple users.

Disadvantages:

- 1.Manual interventions are required between two batches.
- 2.The CPU utilization is low because the time taken in loading and unloading of batches is very high as compared to execution time.

3. Distributed Operating System

- ✓ In a Distributed Operating System, we have various systems and all these systems have their own CPU, main memory, secondary memory, and resources.
- ✓ These systems are connected to each other using a shared communication network. Here, each system can perform its task individually.
- ✓ The best part about these Distributed Operating System is remote access i.e. one user can access the data of the other system and can work accordingly. So, remote access is possible in these distributed Operating Systems.
- ✓ The following image shows the working of a Distributed Operating System.



Advantages:

1. Since the systems are connected with each other so, the failure of one system can't stop the execution of processes because other systems can do the execution.
2. Resources are shared between each other.
3. The load on the host computer gets distributed and this, in turn, increases the efficiency.

Disadvantages:

1. Since the data is shared among all the computers, so to make the data secure and accessible to few computers, you need to put some extra efforts.
2. If there is a problem in the communication network then the whole communication will be broken.

4. Embedded Operating System

- ✓ An Embedded Operating System is designed to perform a specific task for a particular device which is not a computer.
- ✓ For example, the software used in elevators is dedicated to the working of elevators only and nothing else. So, this can be an example of Embedded Operating System. The Embedded Operating System allows the access of device hardware to the software that is running on the top of the Operating System.

Advantages:

1. Since it is dedicated to a particular job, so it is fast.
2. Low cost.
3. These consume less memory and other resources.

Disadvantages:

1. Only one job can be performed.
2. It is difficult to upgrade or is nearly scalable.

5. Real-time Operating System

- ✓ The Real-time Operating Systems are used in the situation where we are dealing with some real-time data. So, as soon as the data comes, the execution of the process should be done and there should be no delay i.e. no buffer delays should be there.
- ✓ Real-time OS is a time-sharing system that is based on the concept of clock interrupt. So, whenever you want to process a large number of request in a very short period of time, then you should use Real-time Operating System.
- ✓ For example, the details of the temperature of the petroleum industry are very crucial and this should be done in real-time and in a very short period of time. A small delay can result in a life-death situation. So, this is done with the help of Real-time Operating System.
- ✓ There are two types of Real-time Operating System:
 - 1.Hard Real-time:** In this type, a small delay can lead to drastic change. So, when the time constraint is very important then we use the Hard Real-time.
 - 2.Soft Real-time:** Here, the time constraint is not that important but here also we are dealing with some real-time data.

Advantages:

1. There is maximum utilization of devices and resources.
2. These systems are almost error-free.

Disadvantages:

1. The algorithms used in Real-time Operating System is very complex.
2. Specific device drivers are used for responding to the interrupts as soon as possible.

Functions of Operating System

- **Processor Management:** An operating system manages the processor's work by allocating various jobs to it and ensuring that each process receives enough time from the processor to function properly.
- **Memory Management:** An operating system manages the allocation and de-allocation of the memory to various processes and ensures that the other process does not consume the memory allocated to one process.
- **Device Management:** There are various input and output devices. An OS controls the working of these input-output devices. It receives the requests from these devices, performs a specific task, and communicates back to the requesting process.

- **File Management:** An operating system keeps track of information regarding the creation, deletion, transfer, copy, and storage of files in an organized way. It also maintains the integrity of the data stored in these files, including the file directory structure, by protecting against unauthorized access.
- **Security:** The operating system provides various techniques which assure the integrity and confidentiality of user data. Following security measures are used to protect user data:
 - Protection against unauthorized access through login.
 - Protection against intrusion by keeping Firewall active.
 - Protecting the system memory against malicious access.
 - Displaying messages related to system vulnerabilities.

- **Error Detection:** From time to time, the operating system checks the system for any external threat or malicious software activity. It also checks the hardware for any type of damage. This process displays several alerts to the user so that the appropriate action can be taken against any damage caused to the system.
- **Job Scheduling:** In a multitasking OS where multiple programs run simultaneously, the operating system determines which applications should run in which order and how time should be allocated to each application.